

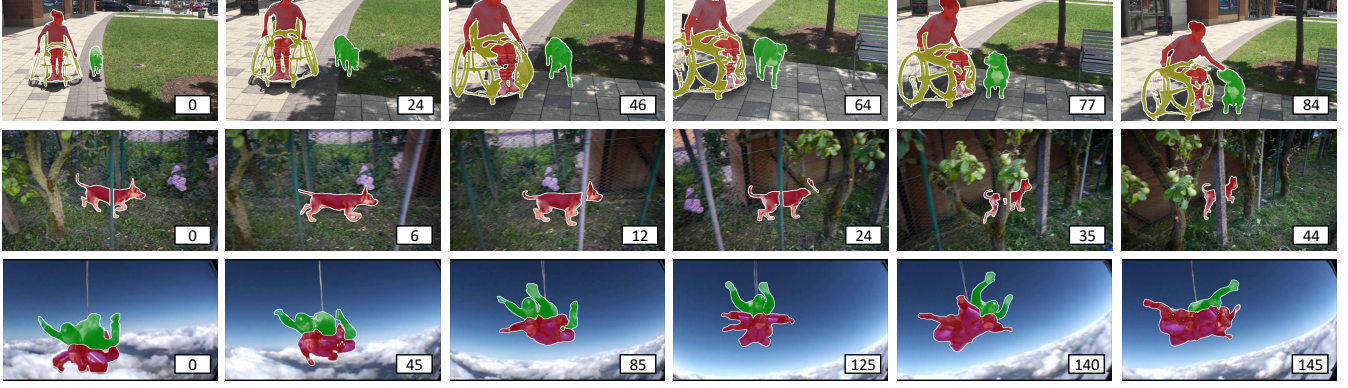


Figure 4: Qualitative results of our method, where frames are sampled at the important moments (e.g. multi-view or occlusion) for each video. From top to bottom, the sequences are “girl-dog” in DAVIS2017, “libby” in DAVIS2016 and “0daaddc9da” in YouTube-VOS, respectively.



Figure 5: Visual comparison with detection based (?) and matching based (?) methods.

α , which controls the operation of the edge weighting, has another improvement of 0.8.

Effectiveness of the temporal graph. Our temporal graph helps to refine the current frame mask result by using the previous frame masks, which can provides the temporal contextual information for better boundary results. As shown in table, the temporal graph takes mask memory for refinement and has the second maximum improvement of 2.1. The hyperparameters λ_1 in Eq. (6), which controls the operation of mask choosing, has another improvement of 0.6.

How to choose the number of graph layer. We further investigate the performances on different iteration number l of graph updating process. Although more graph layer will bring better performance, too much layers will result in over-smoothing problem. We find that the graph with 2-steps updating achieves the best result of 63.1.

Comparison on different training strategy. We also do the experiments on different training strategy to investigate the benefits. Results show that online leaning with the annotations of the first frame brings the improvement of 1.2. Fine-tuning on the augmented train-set of DAVIS2017 makes the improvement of 2.2. And we add YouTube-VOS dataset for pre-training, it has another improvement of 2.9.

Conclusion

In this paper, we propose a spatiotemporal graph neural network (STG-Net) for video object segmentation. By the co-operation of spatial and temporal graph networks, STG-Net has sufficient capacity to reconstruct more detailed masks. Contrasted to the previous detection based approaches uti-

Settings						OL	FT	+ytb	$\mathcal{G}_M$
Mo	SG	α	l	λ_1	TG				
✗	✗	-	-	0.5	✗	✗	✗	✗	53.2
✓	✗	-	-	0.5	✗	✗	✗	✗	54.1
✓	✓	0.5	1	0.5	✗	✗	✗	✗	58.4
✓	✓	0.7	1	0.5	✗	✗	✗	✗	59.2
✓	✓	0.7	1	0.4	✗	✗	✗	✗	59.8
✓	✓	0.7	1	0.4	✓	✗	✗	✗	61.9
✓	✓	0.7	2	0.4	✓	✗	✗	✗	<u>63.1</u>
✓	✓	0.7	3	0.4	✓	✗	✗	✗	62.3
✓	✓	0.7	2	0.4	✓	✓	✗	✗	64.3
✓	✓	0.7	2	0.4	✓	✓	✓	✗	66.5
✓	✓	0.7	2	0.4	✓	✓	✓	✓	69.4

Table 4: Ablation study evaluated on the DAVIS2017 test-dev set. ‘Mo’ means the motion model, ‘SG’ and ‘TG’ mean the spatial and temporal graph. α controls edge weights where $\alpha = 1 - \beta$, and λ_1 controls the mask selection score where $\lambda_1 = 1 - \lambda_2$. l represents the spatial graph iteration number. ‘OL’ indicates online learning and ‘FT’ indicates fine-tuning with lucid (?) augmentation. ‘+ytb’ denotes pre-training on YouTube-VOS dataset.

lizing greed search strategy only in the current frame, the abundant use of both local patch details in spatial graph and time dimension relationships in temporal graph make STG-Net able to obtain a superior representation for the class-agnostic objects segmentation. Extensive experiments demonstrate that our method is robust to challenging scenarios thanks to our spatiotemporal graph, and outperforms state-of-the-art methods on all four benchmarks, even compared to online learning methods.

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